

Linear Algebra and its Applications

Assignment 6

Due: 23 March 2026

Instructions: The solutions to this assignment are to be uploaded on Moodle as a PDF file. For questions 5 and 6 you are expected to produce Python implementation. You may either write all your solutions in $\text{\LaTeX}$ (including the last two questions, don't write the matrix entries) and compile a PDF output or prepare a Jupyter Notebook (answers to first 3 questions written as markdown) and save it as a PDF output).

1. Let $q_1, q_2 \in \mathbb{R}^3$ be orthonormal vectors and let

$$Q = [q_1 \ q_2].$$

- (a) Explain why the matrix QQ^T maps any vector $b \in \mathbb{R}^3$ to the closest vector in the subspace spanned by q_1 and q_2 .
- (b) Describe geometrically what the matrix

$$I - QQ^T$$

does to a vector.

- (c) Suppose $b = p + r$ where p lies in the column space of Q and r is orthogonal to it. Explain how $QQ^T b$ and $(I - QQ^T)b$ recover these components.

2. Let $Q \in \mathbb{R}^{n \times n}$ be an orthogonal matrix.

- (a) Show that for any vector x ,

$$\|Qx\|_2 = \|x\|_2.$$

- (b) Consider the least squares problem

$$\min_x \|Ax - b\|.$$

Explain why replacing A with $Q^T A$ and b with $Q^T b$ does not change the minimizer.

3. Let

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

- (a) Perform the classical Gram–Schmidt process to compute the QR decomposition $A = QR$.
- (b) Verify that

$$Q^T Q = I.$$

- (c) What geometric information about the columns of A is captured by the entries of R ?

4. Consider the matrix

$$A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 + \epsilon \\ 1 & 1 + \epsilon & 1 \end{bmatrix}$$

with

$$\epsilon = 10^{-8}.$$

- (a) Implement the **classical Gram–Schmidt** algorithm.
- (b) Implement the **modified Gram–Schmidt** algorithm.

(c) Compute the matrices Q_{CGS} and Q_{MGS} .

(d) Measure the orthogonality error

$$\|I - Q^T Q\|_F$$

for both methods.

(e) Compare the results and briefly explain which method preserves orthogonality better.

5. Generate a random matrix

$$A \in \mathbb{R}^{200 \times 100}.$$

(a) Compute a QR factorization using your implementation of Modified Gram–Schmidt.

(b) Compute QR using the built-in routine in Python.

(c) Compare the orthogonality error

$$\|I - Q^T Q\|.$$

(d) Solve the least squares problem

$$Ax = b$$

for a random vector b .

(e) Compare the residual norms $\|Ax - b\|$ and comment briefly on the numerical behavior you observe.